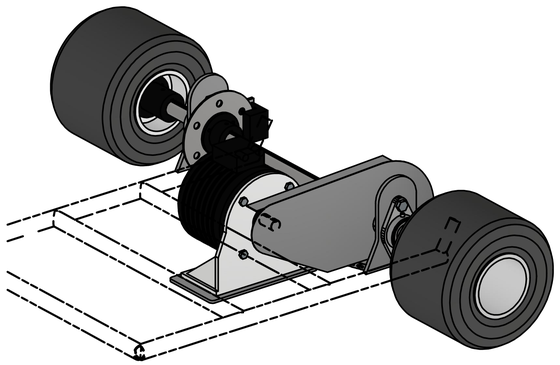
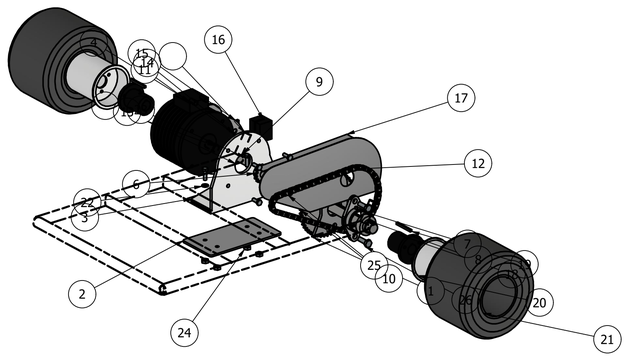
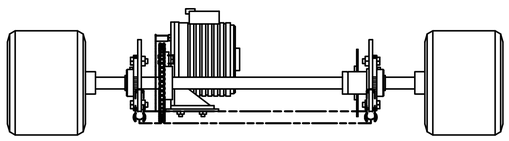
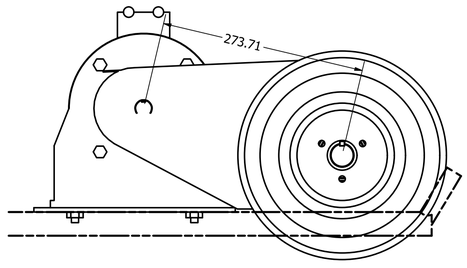




PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
15	1	GK-704	Caliper bracket 1/4 in A36 plate 68 x 105, TIG-welded to the inboard face of the right bearing hanger, 2 x 8.5 for the caliper
16	1	GK-703	Hydraulic kart brake caliper (single piston, rear) envelope + master cylinder + line (GoPowerSports KDMIDSBRKIT)
17	1	GK-705	Chain and sprocket guard, 1.5 mm steel laser cut + bent: outboard side plate + top cover over the tight span, bolted to the motor-mount base (fixings not modelled)
18	2	GK-411	Rear wheel hub 1-1/4 bore x 80 long, 1/4 keyway, 3 holes on 2.5 in BC, 90 flange, 6061-T6 (JOES 25560 class envelope)
19	2	GK-512	Square key 1/4 x 1/4 x 70, 1018 CD (wheel hubs)
20	2	GK-412	Kart rim 5 in dia x 8 in wide, 3 holes on 2.5 in BC, cast aluminium (envelope)
21	2	GK-413	Kart slick 11 x 7.10-5 (rear), envelope
22	4	GK-528	Flat washer DIN 125 M10 (10.5 x 20 x 2), used under the 3/8 slot bolts
23	4	GK-521	Hex head cap screw 3/8-16 x 1-1/4, SAE J429 grade 8, zinc-yellow (motor-mount slot bolts)
24	4	GK-525	Nylon-insert locknut 3/8-16, grade 8
25	4	GK-522	Hex head cap screw 3/8-16 x 1, SAE grade 8 (motor face to mount plate)
26	4	GK-523	Hex head cap screw 1/2-13 x 1-1/2, SAE grade 8 (bearing flange to hanger)
27	4	GK-526	Nylon-insert locknut 1/2-13, grade 8

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	GK-101	Rear-axle bearing hanger, 3/8 in 4130 plate, R25 blends, saddled and TIG-welded to the main rail, 52 bore + 2 x 14
2	1	GK-102	Motor-mount base plate 270 x 142 x 6, A36, welded across the two 1-1/4 cross tubes, 4 x 10.5
3	1	GK-103	Motor mount bracket, 6061-T6 machined from solid: 3/8 in motor plate + 3/8 in base flange with 4 x 24 mm tensioning slots + 2 gussets
4	1	GK-201	Motenergy ME1114 brushless PMSM, 24-72 V, 0.12 Nm/A, 65 Nm peak, 5000 rpm, 7/8 shaft (catalogue envelope)
5	1	GK-513	Square key 3/16 x 3/16 x 19, 1018 CD (motor sprocket)
6	1	GK-401	Sprocket ANSI 40 13T, 7/8 bore, 3/16 key, B-hub 1.563 in (PD 53.07, OD 59.15), steel, Martin 40BS13 7/8
7	2	GK-421	Flange housing UCFL206 2-bolt, cast iron, bolt centres 117 (catalogue dimensions)
8	2	GK-422	Insert ball bearing UC206-20, 1-1/4 bore, set-screw locking, C 19.5 kN, C0 11.3 kN
9	1	GK-301	Rear live axle 1-1/4 in x 40 in, 4140 pre-hard 28-32 HRC, 1/4 x 1/8 full-length keyway, ends chamfered 1x45
10	1	GK-402	Sprocket ANSI 40 50T, 1-1/4 bore + 1/4 keyway, B-hub 3.5 in, 2 set screws (PD 202.26, OD 209.48), steel, Martin 40BS50 1-1/4: keyed straight to the axle (no bolt-on hub)
11	2	GK-511	Square key 1/4 x 1/4 x 25, 1018 CD (axle sprocket hub and brake hub, 2)
12	1	GK-403	Roller chain ANSI 40 single strand, 78 links incl. connecting link (990.6 mm), min tensile 13.9 kN (ANSI), max allowable 3.60 kN (Tsubaki)
13	1	GK-701	Brake-disc hub 1-1/4 bore, 1/4 keyway, 3 x 1/4 on 2.75 in BC (GoPowerSports KD125MB, catalogue-class envelope)
14	1	GK-702	Brake disc 7.1 in (180 mm) x 1/8 in steel, 3-bolt on 2.75 in (GoPowerSports KD711.3)





GOKARTDRIVETRAIN - ELECTRIC KART REAR DRIVETRAIN, 25 MPH (DESIGNED AND SIMULATED, NOT BUILT)
MOTOR MOTENERGY ME1114 PMSM 48 V, 0.12 NM/A, 350 A CONTROLLER = 42 NM; 16S LIFEPO4 40 AH
REDUCTION ANSI 40 CHAIN 13/50 T = 3.846:1, 78 LINKS, CENTRES 285.4 MM; 25 MPH = 2996 RPM MOTOR / 779 RPM AXLE
AXLE 1-1/4 IN 4140 PH LIVE, UC206-20 IN UCFL206 ON 3/8 IN 4130 HANGERS; TYRES 11X7.10-5, R ROLL 137 MM
SIMULATED 0-25 MPH 2.14 S (185 KG), CHAIN SF 8.6 ON MIN. ULTIMATE, 44 % OF ALLOWABLE AT 42 NM
BRAKE 180 MM DISC + HYDRAULIC CALIPER ON THE RIGHT HANGER (127 N M NEEDED); RIMS 5 X 8 IN ON 3 X 2.5 IN BC HUBS
ASSEMBLY PARAMETERS: Drive (AXLE + MOTOR ROTATION), ChainLine (SPROCKET PLANE, Y) AND ChainAdjust (MOTOR SLOT, X)
EXPLODED VIEW = GoKartDrivetrain_Exploded.iam. THREADS COSMETIC. CHASSIS SECTION = REFERENCE (NOT IN PARTS LIST)

DRAWN Jay Patel	10/29/2026	
CHECKED		TITLE
QA		GoKartDrivetrain - general assembly
MFG		
APPROVED		SIZE D | DWG NO GK-DWG-001 | REV
		SCALE 1 / 6 | SHEET 1 OF 1